



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2018-0192



Agricultural Machinery – Rice Combine Harvester ZOOMLION 4LZT-4.0Z

Test Report Number	2018 – 0192
Valid Until	May 22, 2023
Agricultural Machinery	Rice combine harvester
Type	Ride-on, crawler type, sack type, self-propelled
Brand	ZOOMLION
Model	4LZT-4.0Z
Test Requested by	FIRSTECH MACHINERY PHILIPPINES, INC. Unit 1701, One Global Place, 5 th Avenue Corner 25 th Street, Bonifacio Global City, Taguig

SUMMARY

Performance Criteria	Based on PNS/PAES 224:2015	AMTEC Test
Field Efficiency, %, minimum	75.0	81.1
Header Loss, %, maximum	0.5	0.50
Blower/Separation Loss, %, maximum	2.5	2.11
Unthreshed Loss, %, maximum	0.2	0
Scattering Loss, %, maximum	0.3	0
Purity with Fan, %, minimum	97	97.64
Broken Grains, %, maximum	2.0	0.30
Net Cracked Grain, %, maximum	5.0	3.83
Noise Level, db(A), maximum	95*	93.9

*Allowable noise level for four (4) hours of continuous exposure based on Occupational Safety and Health Standards, Department of Labor and Employment, Philippines. 2013.

OBSERVATION/S

1. Manufacturer	ZOOMLION HEAVY MACHINERY CO., LTD. No. 16 E'xi Road, Sanshan Wuhu, Anhui Province, P.R. China
2. Power Rating of Prime Mover	65.6 kW Diesel Engine
3. Operator's Manual	Not Provided
4. Previous Valid Test Reports of Similar Brand and Model	None
5. Handling and stability when machine is working and turning	Machine is stable while working and turning.
6. Manipulating the operating lever	Controls are accessible to the operator.
7. Adjusting and repair of parts	Not observed.
8. Transporting the machine	Not observed.
9. Safety	Warning stickers are available on the machine.
10. Vibration	Not observed.
11. Number of Operators	3
12. Other Remarks	No breakdown or failure of components occurred during field performance test.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	May 23, 2018
Location of Test	Sanggalang, San Antonio, Nueva Ecija
Items Tested	Machine Specifications, Field Capacity, Field Efficiency, Output Capacity, Grain Losses, Grain Quality, Fuel Consumption, Noise Level, and Number of Operators

METHODS OF TEST

The machine was tested based on the Philippine National Standard "PNS/PAES 225:2015 Agricultural Machinery – Rice Combine Harvester – Methods of Test".

DESCRIPTION OF THE MACHINE

Figure 1. Parts of the ZOOMLION 4LZT-4.0ZD Rice Combine Harvester:
(a) Canopy, (b) Grain Tank, (c) Crawler, and (d) Reel assembly,

MACHINE SPECIFICATIONS

ITEM	Manufacturer's Specifications ¹	Verification by AMTEC ²
1 Overall dimensions and weight of rice combine harvester		
1.1 Length, mm	5270	NM
1.2 Width, mm	2500	NM
1.3 Height, mm	2960	NM
1.4 Total weight of the machine, kg	3260	NM
2 Crop condition		
2.1 Variety	ND	“SL7”
2.2 Maximum cutting angle of straw	ND	ND
2.3 Plant height	ND	1104
2.4 Crop moisture content	ND	20.2
3 Harvesting condition		
3.1 No. of cutting rows	ND	ND
3.2 Cutting method	ND	Reciprocating cutting knife
3.3 Cutting width, mm	2000	2000
3.4 Minimum cutting height, mm	ND	152
3.5 Field capacity, ha/h	ND	0.639
4 Starting system	ND	ND
5 Operating condition	ND	ND
6 Harvesting speed, kph	ND	3.94
7 Ground drive	ND	ND

¹Based on machine nameplate;
ND – No Data

²Based on actual measurement and test data;
NM – Not Measured

FIELD TEST RESULTS

ITEM		DATA
1	Crop Condition	
1.1	Variety	“SL7”
1.2	Date of sowing/planting	ND
1.3	Row spacing	Uneven
1.4	Recommended period of maturity	ND
1.5	Average plant height, mm	1104
1.6	Lodging angle of plant	ND
1.7	Plant population/m ²	ND
1.8	Number of tillers/plant	ND
2	Field conditions	
2.1	Location of test field	Sanggalang, San Antonio, Nueva Ecija
2.2	General topography (undulating/leveled)	Leveled
2.3	Area of test field, m ²	2065
2.3.1	Length, m	80.0
2.3.2	Width, m	25.5
2.4	Shape of field	Rectangular
2.5	Soil type	ND
2.6	Soil hardness, kg/cm ²	0.79
3	Test conditions	
3.1	Date of test	May 23, 2018
3.2	Duration of test, min	19.3
3.3	Time lost, min	-
3.4	Operating speed, kph	3.94
3.5	Effective width of cut, mm	2000
3.6	Fuel consumed, ml	2100
3.7	Fuel consumption, L/h	6.37
3.8	Noise level, db(A)	93.9
3.9	Height of cut or height of stubbles, mm	152
3.10	Threshed grain output capacity, kg/h	4816.2

ND – No Data

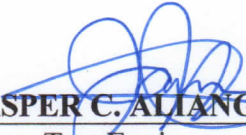
FIELD TEST RESULTS (continuation)

3.11	Actual field capacity	
3.11.1	ha/h	0.639
3.11.2	h/ha	1.566
3.12	Theoretical field capacity	
3.12.1	ha/h	0.788
3.12.2	h/ha	1.269
3.13	Field efficiency, %	81.1
3.14	Number of operators	3 (1 operator and 2 bagger)

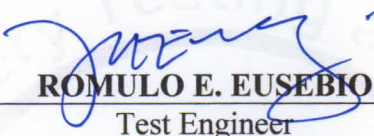
LABORATORY TEST RESULTS

1	Grain-straw ratio	0.58
2	Purity, %	97.64
3	Potential Yield, g/m ²	1257
4	Moisture Content, wet basis, %	20.2
5	Total Losses, %	2.61
5.1	Header Loss, %	0.50
5.2	Blower/Separation Loss, %	2.11
5.3	Unthreshed Loss, %	0
5.4	Scattering Loss, %	0
6	Broken Grains, %	0.30
7	Cracked Grains, %	
7.1	Manually Threshed, %	0
7.2	Machine Threshed, %	3.83
7.3	Net, %	3.83

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APR 29 2019

Date

REMINDER

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